

Programming Fundamental

Lab 12

Task 1: Given a 2D array (matrix) of integers, write a program to find the **sum of all elements** in the array.

Example Input:

```
matrix = [  
    [1, 2, 3],  
    [4, 5, 6],  
    [7, 8, 9]  
]
```

Output:

45

Task 2: Write a program that takes a 2D array and returns its **transpose** (swap rows with columns).

Example Input:

```
matrix = [  
  
    [1, 2, 3],  
  
    [4, 5, 6]  
]
```

Output:

```
[  
    [1, 4],  
    [2, 5],  
    [3, 6]  
]
```

Task 3: Write a program to print the **primary diagonal elements** of a square matrix.

Example Input:

```
plaintext
CopyEdit
matrix = [
    [5, 1, 9],
    [2, 7, 6],
    [3, 4, 8]
]
```

Output:

```
plaintext
CopyEdit
5 7 8
```

Task 4: Given a 2D array and an integer x, write a function to **search for x** in the array and return its position (row, col). If not found, return -1.

Example Input:

```
matrix = [
    [10, 20, 30],
    [40, 50, 60],
    [70, 80, 90]
], x = 50
```

Output:

```
(1, 1)
```

Task 5: Write a function to find the **index of the row** in a 2D array that has the **maximum sum of its elements**.

Example Input:

```
matrix = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
]
```

Output:

```
2
```